



Banking Portfolio Risk Analysis & Financial Modeling End-to-End Financial Modeling Case Study



Project Objective:

Developed an end-to-end financial modeling project to evaluate loan repayment, lending profitability, investment decisions, debt obligations, depreciation schedules, cost of capital, and intrinsic company valuation using Microsoft Excel.

Business Problem:

Banks and financial institutions require structured financial models to evaluate:

*Loan repayment behaviour
Customer profitability
Early loan closure impact
Capital investment decisions
Debt management
Company valuation*

Financial Concepts Covered:

*Time Value of Money
Loan Amortization
Early Loan Settlement
Capital Budgeting
Depreciation
Debt Financing
Cost of Capital
Discounted Cash Flow Valuation
Enterprise Value
Equity Value
Intrinsic Value
Sensitivity Analysis*

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Loan Portfolio Risk & Improving Profitability - CASE STUDY

DATA SET

Loan ID	Customer	Branch	Loan Amount	Interest	Tenure	Start Date
L001	Arjun	Hyderabad	8,00,000	10%	5	Jan-26
L002	Priya	Bangalore	12,00,000	12%	7	Mar-26
L003	Ravi	Chennai	5,00,000	8%	4	Jun-26
L004	Meera	Pune	15,00,000	11%	10	Aug-26

LOAN	L001	1	1	1	1
Year	Opening Loan	Interest (IPMT)	Principal (PPMT)	EMI (PMT)	Closing Loan
1	₹ 8,00,000	₹ 80,000	₹ 1,31,038	₹ 2,11,038	₹ 6,68,962
2	₹ 6,68,962	₹ 66,896	₹ 1,44,142	₹ 2,11,038	₹ 5,24,820
3	₹ 5,24,820	₹ 52,482	₹ 1,58,556	₹ 2,11,038	₹ 3,66,264
4	₹ 3,66,264	₹ 36,626	₹ 1,74,412	₹ 2,11,038	₹ 1,91,853
5	₹ 1,91,853	₹ 19,185	₹ 1,91,853	₹ 2,11,038	₹ 0
		₹ 2,55,190			

LOAN	L002	1	1	1	1
Year	Opening Loan	Interest (IPMT)	Principal (PPMT)	EMI (PMT)	Closing Loan
1	₹ 12,00,000	₹ 1,44,000	₹ 1,18,941	₹ 2,62,941	₹ 10,81,059
2	₹ 10,81,059	₹ 1,29,727	₹ 1,33,214	₹ 2,62,941	₹ 9,47,844
3	₹ 9,47,844	₹ 1,13,741	₹ 1,49,200	₹ 2,62,941	₹ 7,98,645
4	₹ 7,98,645	₹ 95,837	₹ 1,67,104	₹ 2,62,941	₹ 6,31,541
5	₹ 6,31,541	₹ 75,785	₹ 1,87,156	₹ 2,62,941	₹ 4,44,384
6	₹ 4,44,384	₹ 53,326	₹ 2,09,615	₹ 2,62,941	₹ 2,34,769
7	₹ 2,34,769	₹ 28,172	₹ 2,34,769	₹ 2,62,941	0.00
		₹ 6,40,589			

LOAN	L003	1	1	1	1
Year	Opening Loan	Interest (IPMT)	Principal (PPMT)	EMI (PMT)	Closing Loan
1	₹ 5,00,000	₹ 40,000	₹ 1,10,960	₹ 1,50,960	₹ 3,89,040
2	₹ 3,89,040	₹ 31,123	₹ 1,19,837	₹ 1,50,960	₹ 2,69,202
3	₹ 2,69,202	₹ 21,536	₹ 1,29,424	₹ 1,50,960	₹ 1,39,778
4	₹ 1,39,778	₹ 11,182	₹ 1,39,778	₹ 1,50,960	₹ 0
		₹ 1,03,842			

LOAN	L004	1	1	1	1
Year	Opening Loan	Interest (IPMT)	Principal (PPMT)	EMI (PMT)	Closing Loan
1	₹ 15,00,000	₹ 1,65,000	₹ 89,702	₹ 2,54,702	₹ 14,10,298
2	₹ 14,10,298	₹ 1,55,133	₹ 99,569	₹ 2,54,702	₹ 13,10,728
3	₹ 13,10,728	₹ 1,44,180	₹ 1,10,522	₹ 2,54,702	₹ 12,00,206
4	₹ 12,00,206	₹ 1,32,023	₹ 1,22,679	₹ 2,54,702	₹ 10,77,527
5	₹ 10,77,527	₹ 1,18,528	₹ 1,36,174	₹ 2,54,702	₹ 9,41,353
6	₹ 9,41,353	₹ 1,03,549	₹ 1,51,153	₹ 2,54,702	₹ 7,90,200
7	₹ 7,90,200	₹ 86,922	₹ 1,67,780	₹ 2,54,702	₹ 6,22,419
8	₹ 6,22,419	₹ 68,466	₹ 1,86,236	₹ 2,54,702	₹ 4,36,183
9	₹ 4,36,183	₹ 47,980	₹ 2,06,722	₹ 2,54,702	₹ 2,29,461
10	₹ 2,29,461	₹ 25,241	₹ 2,29,461	₹ 2,54,702	₹ 0
		₹ 10,47,021			

	1	1	1
Loan ID	Cumulative Interest	Cumulative Principal	Total Cumulative Repayment
L001	₹ 2,55,190	₹ 8,00,000	₹ 10,55,190
L002	₹ 6,40,589	₹ 12,00,000	₹ 18,40,589
L003	₹ 1,03,842	₹ 5,00,000	₹ 6,03,842
L004	₹ 10,47,021	₹ 15,00,000	₹ 25,47,021

Create Assumptions

Input	Value
Customer Name	Arjun Kumar
Original Loan Amount	₹ 8,00,000
Interest Rate	10%
Original Loan Tenure	10
Loan Closed After	3
Foreclosure Charge	2%

KPI	Value
Original Loan	₹ 8,00,000
Loan Closed After	3
Outstanding Loan	₹ 5,24,820
Total Interest Earned	₹ 2,55,190
Interest Lost	₹ 55,812
Foreclosure Fee	₹ 10,496
Net Loss to Bank	₹ 45,315

Dynamic Loan Amortization Schedule with scenario-based repayment analysis.

INVESTMENT DECISION ANALYSIS

Year	Corporate (cr)	Housing (cr)
0	-50	-50
1	18	12
2	20	15
3	22	18
4	25	22
5	28	26
	113	93

Input	Project A	Project B
Initial Investment	₹ 50	₹ 50
Discount Rate (WACC)	10%	10%
Reinvestment Rate	12%	12%

Analysis	Project A	Project B
Present Value (PV)	31	31
Future Value (FV)	₹ 80.53	₹ 80.53
Net Present Value (NPV)	30.80	₹ 16.36
Internal Rate of Return (IRR)	32%	21%
Modified IRR (MIRR)	23%	18%
Required Rate of Return (RRI)	18%	13%

Conclusion: Project A is recommended over Project B as it demonstrates superior performance across key capital budgeting metrics such as NPV, IRR, MIRR, and RRI, indicating stronger profitability and investment potential.

Depreciation Schedule

Asset	Cost	Life	Salvage
ATM	25,00,000	6	1,00,000
Server	40,00,000	5	2,00,000
Laptop	1,20,000	4	10,000

Straight Line Depreciation:

	1	1	1	1	1	1	
PARTICULARS	2026	2027	2028	2029	2030	2031	
ATM	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000	₹ 4,00,000	₹ 24,00,000
Server	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000	₹ 7,60,000		₹ 38,00,000
Laptop	₹ 27,500	₹ 27,500	₹ 27,500	₹ 27,500			₹ 1,10,000

Fixed Declining

	1	2	3	4	5	6	
PARTICULARS	2026	2027	2028	2029	2030	2031	
ATM	₹ 10,37,500	₹ 6,06,938	₹ 3,55,058	₹ 2,07,709	₹ 1,21,510	₹ 71,083	₹ 23,99,798
Server	₹ 18,04,000	₹ 9,90,396	₹ 5,43,727	₹ 2,98,506	₹ 1,63,880		₹ 38,00,510
Laptop	₹ 55,560	₹ 29,836	₹ 16,022	₹ 8,604			₹ 1,10,021

Double Declining Depreciation:

	1	2	3	4	5	6	
PARTICULARS	2026	2027	2028	2029	2030	2031	
ATM	₹ 8,33,333	₹ 5,55,556	₹ 3,70,370	₹ 2,46,914	₹ 1,64,609	₹ 1,09,739	₹ 22,80,521
Server	₹ 16,00,000	₹ 9,60,000	₹ 5,76,000	₹ 3,45,600	₹ 2,07,360		₹ 36,88,960
Laptop	₹ 60,000	₹ 30,000	₹ 15,000	₹ 5,000			₹ 1,10,000

Sun of the year digits

	1	2	3	4	5	6	
PARTICULARS	2026	2027	2028	2029	2030	2031	
ATM	₹ 6,85,714	₹ 5,71,429	₹ 4,57,143	₹ 3,42,857	₹ 2,28,571	₹ 1,14,286	₹ 24,00,000
Server	₹ 12,66,667	₹ 10,13,333	₹ 7,60,000	₹ 5,06,667	₹ 2,53,333		₹ 38,00,000
Laptop	₹ 44,000	₹ 33,000	₹ 22,000	₹ 11,000			₹ 1,10,000

Variable Declining Depreication	FACTOR						
	0	1	2	3	4	5	6
PARTICULARS	2026	2027	2028	2029	2030	2031	
ATM	₹ 8,33,333	₹ 5,55,556	₹ 3,70,370	₹ 2,46,914	₹ 1,64,609	₹ 1,09,739	₹ 22,80,521
Server	₹ 16,00,000	₹ 9,60,000	₹ 5,76,000	₹ 3,45,600	₹ 3,18,400		₹ 38,00,000
Laptop	₹ 60,000	₹ 30,000	₹ 15,000	₹ 5,000			₹ 1,10,000

Debt Schedule: Bank Debt

Input	Value (cr)
Loan Amount	₹ 1,00,00,00,000
Interest Rate	8%
Loan Term	5
Repayment Type	Equal Annual Payments

	1	1	1	1	1
Year	Opening Debt	Interest Expense	Principal Repaid	Total Payment	Closing Debt
1	₹ 1,00,00,00,000	₹ 8,00,00,000	₹ 17,04,56,455	₹ 25,04,56,455	₹ 82,95,43,545
2	₹ 82,95,43,545	₹ 6,63,63,484	₹ 18,40,92,971	₹ 25,04,56,455	₹ 64,54,50,575
3	₹ 64,54,50,575	₹ 5,16,36,046	₹ 19,88,20,409	₹ 25,04,56,455	₹ 44,66,30,166
4	₹ 44,66,30,166	₹ 3,57,30,413	₹ 21,47,26,041	₹ 25,04,56,455	₹ 23,19,04,125
5	₹ 23,19,04,125	₹ 1,85,52,330	₹ 23,19,04,125	₹ 25,04,56,455	₹ 0

WACC & DCF

Input	Value (cr)
Market Value of Debt	₹ 250
Market Value of Equity	₹ 750
Cost of Debt	8%
Cost of Equity	15%
Corporate Tax Rate	25%

Assumptions:	
Revenue	10%
Operating cost	7%
Terminal Rate	3%

Cost of Debt	
Pre Tax	8%
Tax Rate	25%
Post Tax Debt	6.00%

WACC	
Weight of Debt	25%
Debt	6%
Weight of Equity	75%
Equity	15%
Cost of Capital	12.75%

Capital Structure:	
Debt	₹ 250
Equity	₹ 750
Total	₹ 1,000

Weight of Debt	25%
Weight of Equity	75%

WACC represents the minimum return required by investors and lenders. Any *investment project with a return higher than the WACC* is

	0	1	2	3	4	5
Particulars (cr)	2026A	2027F	2028F	2029F	2030F	2031F
Revene	₹ 120	₹ 132	₹ 145	₹ 160	₹ 176	₹ 193
Operating Cost	₹ 71	₹ 76	₹ 81	₹ 87	₹ 93	₹ 100
EBIT	₹ 49	₹ 56	₹ 64	₹ 73	₹ 83	₹ 94
Tax	25%	25%	25%	25%	25%	25%
NOPAT (EBIT*(1-Tax))	₹ 37	₹ 42	₹ 48	₹ 55	₹ 62	₹ 70
Dep	₹ 6	₹ 7	₹ 8	₹ 9	₹ 10	₹ 11
Capex	₹ 10	₹ 11	₹ 12	₹ 13	₹ 14	₹ 15
Working Capital	₹ 3	₹ 5	₹ 7	₹ 9	₹ 11	₹ 13
Change in WC		₹ 2	₹ 2	₹ 2	₹ 2	₹ 2
Reinvestment Amt	₹ 10	₹ 13	₹ 14	₹ 15	₹ 16	₹ 17
FCFF	₹ 33	₹ 36	₹ 42	₹ 49	₹ 56	₹ 64
Discount Rate (WACC)	13%					
PV of FCFF	₹ 32.75	₹ 29.05	₹ 25.76	₹ 22.85	₹ 20.26	₹ 17.97

Terminal Value:	
FCFF _(n+1)	₹ 18.51
WACC	12.75%
Terminal Rate	3%
Terminal Rate	₹ 189.87
Year	5
PV of Terminal Vali	₹ 104.20

Enterprise Value:	
PV of FCFF	₹ 115.90
PV of Terminal Valu	₹ 104.20
Enterprise Value	₹ 220.10
Cash	30
Debt	10
Equity Value	₹ 240.10
No of Shares	100
Intrinsic Value per :	₹ 2.40

Sensitivity Analysis				
WACC				
	7%	8%	9%	10%
	2.77	2.68	2.59	2.51
	2.90	2.8	2.71	2.62
Terminal Rate	4%	2.95	2.85	2.76
	5%	3.13	3.03	2.92

